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Chapter 2

Summarizing and Visualizing Data

"A picture is worth a thousand words."— Frederick Barnard

2.1 Introduction

In the modern era, statisticians and analysts are often faced with large amounts of data, which creates the need to summarize and visualize the data before analysis. Therefore, methods are required for creating easily digestible summaries and visual representations of data. The creation of easily digestible summaries and visual representations enables the analyst to obtain a better idea of the broad patterns in the data. This chapter will introduce algorithms for summarizing and visualizing data. The discussions in these chapters primarily address two issues:

- *Summarizing data:* Data summarization discusses the generation of key statistical properties of the data, such as its measures of *central tendency* or measures of *dispersion*. Measures of central tendency try to quantify measures of central location with respect to the data values, whereas measures of dispersion quantify how much the data is spread around these central locations. Other forms of data summarization include measures of association between numeric variables, or the generation of contingency tables for categorical data.
- *Data visualization:* The analyst can often obtain a better understanding of the data with the use of data visualization techniques. Examples of such visualization techniques include the construction of histograms, scatter plots, and bar charts. Visualization techniques often help the analyst get a better understanding of the details of how the data is distributed (e.g., regions of high frequency and low frequency). In some cases, this type of understanding can provide insights about the type of model that will work best in a particular machine learning application.

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